MIL-I-50406 (MU)
1 April 1970
SUPERSEDING
PA-PD-2622A
12 April 1968

MILITARY SPECIFICATION

Initiator Assembly for Primer, Electric XM91E1 and XM117 Loading, Assembling and Packing

1. SCOPE

1.1 This Specification covers the loading, assembling and packing of the initiator assembly for use in two types of primers designated as Primer, Electric, XM91E1 and XM117.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-650	-	Explosive: Sampling, Inspection and Testing

FSC: 1320

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DRAWINGS

ORDNANCE

9204185	-	Initiator Sub-Assembly
9204186	-	Initiator Assembly
9204226	-	Box, Setup, Packing, Ammunition Inner for Initiator Assembly
9204227	-	Box, Setup, Packing, Ammunition Outer for Initiator Assembly
9204228	-	Box, Packing, Ammunition for Initiator Assembly
7986940	-	Firing Probe Assembly

PUBLICATIONS

ORDNANCE CORPS

ET - 9204186 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initiator Sub-assembly (see dwg. 9204185.)

3.1.1 The welds of the bridge ribbon shall be capable of supporting the weight of the entire subassembly.

3.2 Initiator Assembly (see dwg. 9204186).

3.2.1 Circuit Continuity.-The electrical resistance of the Initiator Assembly shall not be greater than 0.40 ohm or less than 0.20 ohm.

3.2.2 Non-functioning.-The initiator assembly shall not function when energized with a direct current (DC) of 0.3 amperes for 0.3 seconds.

3.2.3 Functioning

3.2.3.1 Regulated Power Supply.-The initiator assembly shall function properly (see 6.5) within 16 milliseconds when energized with a 24 volt DC power supply with current regulated to plus or minus one percent.

3.2.4 Inner sleeve damage.-At completion of the continuity test, the inner sleeve shall not be distorted or damaged.

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3.3 Ignition Charge Composition (dry).-The igniter composition shall consist of the following ingredients and percentages prior to the addition of Egyptian Lacquer.

Ingredient	Percent by Weight
Charcoal	20 ± 3
Lead Thiocyanate	35.5 ± 3
Potassium Chlorate	44.5 ± 3

The above composition shall be defined as a dry mix. A wet mix shall be defined as the above composition with the addition of Egyptian Lacquer. The grades, classes and qualified manufacturers of the ingredients shall be as specified in drawing 9204186.

3.4 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of chips, cracks, dirt, grease, flux, rust, corrosion products and other foreign matter. The cleaning method shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. All assemblies shall be fabricated such that no components are free or loose.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items is submitted to the Government for acceptance, the contractor shall supply the following information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

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a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Sustantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Examination and ballistic tests

4.2.1.1 Submission.-The contractor shall submit a first article sample, as designated by the contracting officer, for evaluation in accordance with the provisions of 4.2.1.2. The first article sample shall consist of 50 initiator sub-assemblies and 50 initiator assemblies which have been produced by the contractor or furnished by a supplier, and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. (A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government).

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4.2.1.2 Inspections to be performed.-Parts will be subjected by the Government to any or all of the examinations or tests specified in 4.3.3 and 4.3.4 of this specification, and any or all requirements of the applicable drawings. All tests shall be performed as specified in 4.4.

4.2.2 Rejection.-If any part fails to comply with the requirements for the inspections performed in accordance with 4.2.1.2, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample, or portion thereof. Until the initial production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production, unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process in accordance with the same drawings, same drawing revisions, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required shall be in accordance with MIL-STD-1168. Each lot shall contain ignition charge mix produced containing constituent materials from one lot (see 6.4).

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4.3.2 Batch formation.-A batch of initiator assemblies shall be defined as those initiator assemblies which have been loaded with igniter charge made from one homogeneous mixing operation. The batches used to form each lot shall be identified so that any testing which requires samples from each batch can be facilitated.

4.3.3 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defect (and major defect, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.5.

4.3.3.1 Initiator Sub-assembly prior to assembly of outer cup (see dwg. 9204185 covering a detail of dwg. 9204186).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Evidence of adhesive spillage on detent surface	Visual	01001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	01002

4.3.3.2 Initiator Sub-assembly (see dwg. 9204185 covering a detail of dwg. 9204186).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	100 percent inspection		
101.	Bridge ribbon broken or multiple bridge ribbon	Visual	02001
102.	Hole in outer cup not aligned with hole in front support	Visual	02002

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (See 3.4).....	Visual	02003

4.3.3.3 Initiator assembly (see dwg. 9204186).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	100 percent inspection		
101.	Ignition charge missing.....	Visual	03001
102.	Ignition charge in slotted area above flush of either front support or outer cup--whichever is higher (see fig. 1).....	Visual	03002
Minor:	AQL 1.0 percent		
201.	Evidence of adhesive on perforated end of outer cup.....	Visual	03003
202.	Evidence of poor workmanship (see 3.4).....	Visual	03004

4.3.3.4 Packaging and Packing for Initiator Assembly

4.3.3.4.1 Box, Set-up, Inner (see dwg. 9204226).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Contents of inner box loose.....	Manual	04001
Minor:	AQL 0.65 percent		
201.	Incorrect marking.....	Visual	04002

4.3.3.4.2 Box, Set-up, Outer (see dwg. 9204227).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Sealing strip incomplete or otherwise fails to seal outer set up box completely.....	Visual	04003

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102.	Contents of outer box loose.....	Manual	04004
103.	Barrier bag not sealed.....	Visual-Manual	04005
Minor:	AQL 0.65 percent		
201.	Incorrect marking on box and bag.	Visual	04006
202.	Evidence of poor workmanship (see 3.4).....	Visual	04012

4.3.3.4.3 Box, Packing for Initiator Assembly (see dwg. 9204228).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Packing box damaged contents exposed.....	Visual	04007
102.	Strapping missing, broken or loose.....	Visual	04008
103.	Box improperly sealed.....	Visual	04009
104.	Contents loose.....	Visual	04010
Minor:	AQL 0.65 percent		
201.	Incorrect marking.....	Visual	04011
202.	Evidence of poor workmanship (see 3.4).....	Visual	04013

4.3.4 Testing

4.3.4.1 Initiator Sub-assembly

4.3.4.1.1 Strength of bridge ribbon weld, (see 3.1.1) Major defect, Code No. 05001.-The bridge ribbon weld of the initiator sub-assembly shall support the weight of the subassembly without breaking when tested in accordance with paragraph 4.4.1.1. This test shall be conducted 100 percent. Any subassembly failing to comply with the requirement in 3.1.1 shall be classed defective and removed from the lot.

4.3.4.2 Initiator assembly.

4.3.4.2.1 Circuit Continuity, (see 3.2.1) Major defect, Code No. 06001.-This test shall be conducted 100 percent. Any initiator assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2.1 using equipment in accordance with 4.3.5. The test shall be performed prior to the non-functioning and functioning tests.

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4.3.4.2.2 Non-functioning, Electrical (see 3.2.2), Major defect, code No. 07001.-Initiator assemblies shall be selected from each lot according to MIL-STD-105 with an AQL of 0.40 percent using a single sampling plan with normal inspection. The test samples shall be divided as equally as possible among the batches. The assemblies shall be subjected to the test specified in 4.4.2.2 using equipment in accordance with 4.3.5. Each sample shall be observed for compliance with the applicable requirement. If a lot fails, it shall be retested in accordance with 4.3.4.2.5. Samples used for this test may also be used for the functioning tests. Once the initiator has been checked for non-functioning and is to be used as a sample for the functioning test, it is to be tested for functioning prior to its removal from the firing pin.

4.3.4.2.3 Functioning, Electrical

4.3.4.2.3.1 Regulated DC Power Supply (see 3.2.3.1), Major defect, Code No. 08001.-Initiator assemblies shall be selected from each lot according to MIL-STD-105 with an AQL of 0.40 percent using a single sampling plan with normal inspection. The samples shall be divided as equally as possible among the batches and subjected to the test described in 4.4.2.3.1, using equipment in accordance with 4.3.5. Each sample shall be observed for compliance with the applicable requirement (see 6.5). Samples used for the non-functioning test may also be used for the functioning tests, provided the initiator is not removed from the test firing pin after the non-functioning and before the functioning test. If a lot fails it shall be retested in accordance with 4.3.4.2.5.

4.3.4.2.4 Inner sleeve damage (3.2.4), Major defect, Code No. 09001.-Initiator assemblies shall be randomly selected from each lot (after being subjected to the circuit continuity test of 4.4.2.1) according to MIL-STD-105, with an AQL of 0.40 percent using a single sampling plan with normal inspection and examined for evidence of distortion or damage of the inner sleeve according to 4.4.2.4 using equipment in accordance with 4.3.5. If the sample fails the applicable requirement, the lot shall be retested in accordance with 4.3.4.2.5. Samples used for the inspection may be used for the functioning and non-functioning tests.

4.3.4.2.5 Retest for non-functioning, functioning and inner sleeve damage.-If a lot is rejected for non-functioning, functioning, or inner sleeve damage, each batch in the lot shall be tested for acceptance in accordance with MIL-STD-105, using an AQL of 0.40, for a double sampling plan with reduced inspection. Results shall be reported as required by 6.6.

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4.3.4.3 Ignition Charge Composition

4.3.4.3.1 Dry mix, (see 3.3) Major defect, Code No. 10001.-A 15 gram sample taken from one batch from each lot shall be analyzed in accordance with 4.4.3. Failure of the dry mix to comply with the applicable requirement shall be cause for rejection of the lot.

4.3.4.3.2 Lacquer solids, Informational.-Each lot of lacquer shall be analyzed for solids content in accordance with 4.4.3.1 for information. The results shall be reported as specified in 6.6 (See 6.4).

4.3.5 Inspection Equipment.-Equipment Tabulation Number ET-9204186 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.5.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.5.2 Government use of contractor's inspection equipment.-The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures.

4.4.1 Initiator Sub-assembly

4.4.1.1 Strength of bridge ribbon weld.-This test shall be conducted using the test equipment and operating procedures listed in the applicable Inspection Equipment Drawing.

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4.4.2 Initiator Assembly

4.4.2.1 Circuit Continuity.-The initiator assembly shall be tested for electrical resistance by applying a direct current of 10 milliamperes max. using a probe through the outer cup hole to the inner sleeve as one contact point. The outer cup shall be used as the second contact point. Assure that the probe and all initiator parts are free of dirt, fingerprints, moisture, grease, etc before attempting this test.

4.4.2.2 Non-functioning.-The initiator assembly shall be tested for non-functioning by applying a direct current of 0.3 amperes for 0.3 seconds through a simulated firing pin (see 4.3.5) across the bridge ribbon. Observation shall be made for functioning.

4.4.2.3 Functioning

4.4.2.3.1 Regulated DC Power Supply.-A simulated firing probe shall be inserted into the base of the initiator so that contact is made between the tip of the probe and the inner sleeve of the initiator and between the outer cup of the initiator and ground of probe. Circuit continuity shall be checked for each unit prior to testing. The initiator shall be actuated with a 24 volt DC power supply with the output regulated to within plus or minus one percent. Observation shall be made for compliance with 3.2.3.1. If the resistance of an initiator is outside the acceptable tolerance limits, the resistance shall be recorded and an attempt to fire the initiator shall be made, the firing time recorded, and the results forwarded for information as required in 6.6. The defective sample shall not be used for acceptance of the initiator lot based on the functioning requirement. The defective sample shall be replaced and the new sample tested. If more than 5% of the samples selected for functioning are found to have resistances outside the tolerance limit, the entire lot shall be rechecked 100% for continuity before being retested for functioning.

4.4.2.4 Inner sleeve damage.-To perform this examination, it is necessary to puncture the cup base and push in the perforated leaves.

4.4.3 Ignition charge composition.-The following test methods shall be used to determine the chemical composition of the dry ignition charge mix. The percent potassium chlorate shall be computed by difference.

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4.4.3.1 Lacquer solids.-Approximately 1.5 grams of the sample shall be transferred to a tared medium porosity filtering crucible and allowed to air dry for about an hour. The crucible and contents shall be placed in a vacuum oven for two hours maintained at a temperature of 55 plus or minus 2 degrees Centigrade. The crucible and contents shall be cooled in a desiccator and weighed. The procedure shall be repeated until the change in weight does not exceed 0.5 milligram. The percent lacquer solids shall be calculated as follows:

$$\text{Percent lacquer solids} = 100\ \frac{(A - B)}{A}$$

Where:

A = Initial weight of sample
B = Weight of the dried sample

4.4.3.2 Charcoal.-Approximately 1.5 grams of the sample shall be transferred to a tared medium porosity filering crucible. The contents of the crucible shall be extracted with ten 25 ml. portions of acetone saturated with potassium chlorate and lead thiocyanate. The crucible and contents shall be aspirated until the odor of acetone is no longer perceptible and then dried in a vacuum oven maintained at 55 plus or minus 2 degrees Centigrade for two hours. The crucible and contents shall be cooled in a desiccator and weighed. Using a Fisher high form filtrater or equivalent extractor the sample shall be extracted with ten 25 ml. portions of a 5 percent potassium nitrate solution followed with 50 ml. of distilled water. A glass rod shall be used to break up any lumps in the sample during the extraction process, the filtrate shall be reserved for the determination of lead thiocyanate. The crucible and contents shall be aspirated for several minutes, then placed in an oven maintained at 100 plus or minus 5 degrees Centigrade for two hours. The crucible and contents shall be cooled in a desiccator and weighed. The percent charcoal shall be calculated as follows:

$$\text{Percent charcoal} = \frac{100\ A}{B}$$

Where:

A = the weight of sample after acetone and 5 percent potassium nitrate solution extraction.
B = the weight of the sample after acetone extraction

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4.4.3.3 Lead thiocyanate.-Exactly 35 ml. of standardized 0.1 normal (N) silver nitrate shall be added to the filrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml. of 3N nitric acid (nitric acid used should be water-white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml. of a saturated solution of ferric ammonium sulfate shall be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent lead thiocyanate} = \frac{16.17\ (AE - CD)}{B}$$

Where:

A = Volume of silver nitrate solution used.
B = Weight of the sample after acetone extraction (see 4.4.3.2).
C = Volume of the potassium thiocyanate solution used.
D = Normality of potassium thiocyanate solution.
E = Normality of the silver nitrate solution.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-Initiator Assemblies shall be preserved and packaged as specified in dwg. 9204226 and 9204227.

5.2 Packing

5.2.1 Level C.-Initiator Assembly packaging shall be as specified in drawing 9204228.

5.3 Marking.-Initiator Assembly packaging and packing shall be marked in accordance with drawing 9204226, 9204227, and 9204228.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Ammunition data cards shall be prepared for each lot in accordance with MIL-STD-1167.
c. Provisions for submission of first article samples.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this Specification. They should also be cited as references on drawing of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA-1010) the numbers of drawings of Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the ET by the following codes:

CD - Contractor design on controlled format and/or commercial equipment.
MD - Army design, mandatory for use.

6.3.3 Army design.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format that the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reporducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5 digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the ET, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND12. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Egyptian lacquer.-A lot of Egyptian lacquer shall be considered to be one shipment of homogeneous material from the same manufacturer.

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6.5 Proper functioning.-The initiator shall be considered to have functioned properly if the ignition mix is completely consumed within the required ignition delay time.

6.6 Reporting Test Results.-The results of all informational testing and retesting shall be forwarded to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-ND2, (one copy).

6.7 Submission of first article sample.-Instructions as to the location for the evaluation of first article samples shall be obtained from the contracting officer, upon request. All inquiries should be forwarded to:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NK1
Dover, New Jersey 07801

6.8 Check test for possible deterioration (applicable to initiator assembly storage).-If the total elapsed time between original acceptance of a lot of initiator assemblies and the assembly of that lot into ammunition components with which it is to be issued exceeds two years, or if the initiators have been subjected to adverse conditions at any time since previous tests, the initiator lot shall be subjected to, and must satisfactorily pass the following check test within six months of time of assembly.

6.8.1 Electrical resistance.-Each initiator shall be tested as specified in 4.4.2.1. If any initiator fails to comply with the applicable requirement, it shall be removed from the lot.

6.8.2 Non-functioning, Electrical.-The sampling plan for this test shall be as specified in Table I with the method of testing in accordance with 4.4.2.2. Any initiator lot which fails to comply with applicable requirements shall be classed defective.

6.8.3 Functioning, Electrical.-The sampling plan for this test shall be in accordance with Table I. Any lot which fails to comply with requirements shall be classed defective. The test shall be performed as specified in 4.4.2.3.

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TABLE I

Sample	Sample Size	Size	Acceptance Number	Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

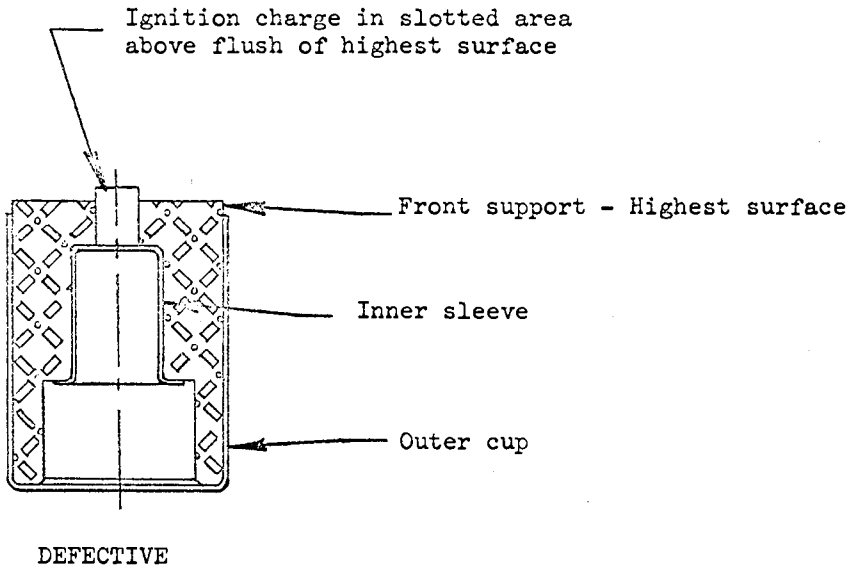
*No acceptance on this sample.

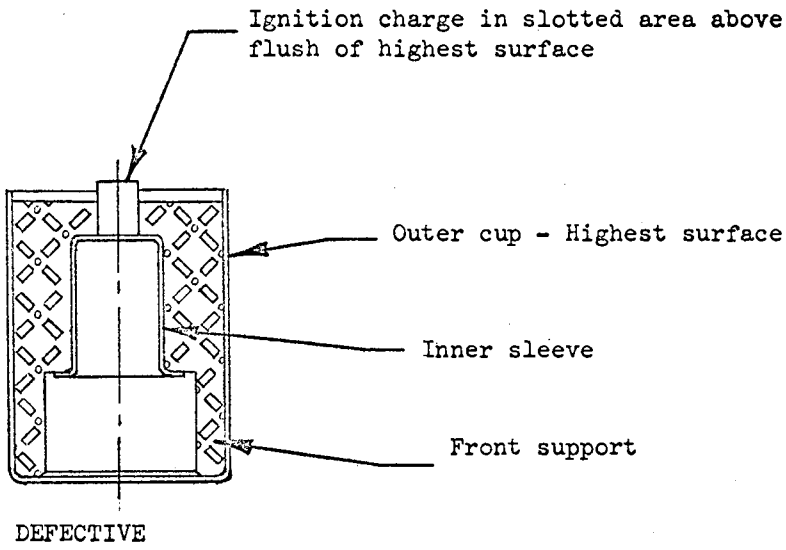
Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Proj. No. 1320-A-157

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EXAMPLES OF DEFECTIVES

FIGURE I

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